

SSL Project: Mini Game Engine

1 Login

- Allow multiple attempts upon wrong password
- Username should not be empty
- Passwords are hashed before storage
- Hashing username to deal with special characters such as \$
- Both players may not have the same username

2 Sign-up

- Ask to confirm password
- Allow multiple attempts if password and confirm password inputs do not match
- Username should not be empty

3 Game Menu

- Settings button for general changes
- Menu with buttons to select game

4 Games

4.1 Tic-Tac-Toe

- Played on a 10×10 board; players need 5 marks in a row to win
- Players alternate placing X or O in any empty cell
- Winning line detection

4.2 Othello

- Allow changing board color (via a settings button)
- Allow selection of token color (via a settings button)
- Allow for various board sizes (including default 8×8)
- Button to show rules
- Game log showing move history and invalid move notifications
- Wipeout status special effects
- Disc flipping animation

4.3 Connect 4

- Highlight column upon mouse hover
- Settings customizations similar to Othello
- Button to show rules

5 Leaderboard Display

- Sorting based on:
 - Game name
 - User
 - Number of wins
 - Number of losses
 - Win/loss ratio
- Priority order for resolving ties
- Option for reverse sorting

6 Visualization

- Pie charts showing most played games of logged-in users

7 Post-game Loop

- **Play another game:** Start a new game of the same type
- **Return to menu:** Go back to the game menu
- **Quit:** Terminate the program